Course notes

Addition and double-angle formulas

Property · Addition formulas. For all real numbers a and b:

$$\cos(a-b)=\cos a\cos b+\sin a\sin b, \qquad \cos(a+b)=\cos a\cos b-\sin a\sin b,$$

$$\sin(a-b)=\sin a\cos b-\cos a\sin b, \qquad \sin(a+b)=\sin a\cos b+\cos a\sin b.$$

Proof. In a direct orthonormal frame, let $\vec{p}\begin{pmatrix}\cos a\\ \sin a\end{pmatrix}$ and $\vec{q}\begin{pmatrix}\cos b\\ \sin b\end{pmatrix}$: these vectors are unit vectors and $(\vec{u},\vec{p})\equiv a$, $(\vec{u},\vec{q})\equiv b \pmod{2\pi}$. By Chasles' relation, $(\vec{q},\vec{p})\equiv a-b \pmod{2\pi}$. We compute $\vec{q}\cdot\vec{p}$ in two ways:

$$\vec{q}\cdot\vec{p}=\left\|\vec{q}\right\|\left\|\vec{p}\right\|\cos(a-b)=\cos(a-b) \quad \text{and} \quad \vec{q}\cdot\vec{p}=\cos b\cos a+\sin b\sin a.$$

This proves the first formula. Replacing b by $-b$ and using $\cos(-b)=\cos b$, $\sin(-b)=-\sin b$, we obtain $\cos(a+b)$. Finally $\sin(a+b)=\cos\left(\frac{\pi}{2}-a-b\right)=\cos\left(\left(\frac{\pi}{2}-a\right)-b\right)$, which the first formula expands into $\sin a\cos b+\cos a\sin b$; we deduce $\sin(a-b)$ by replacing b by $-b$. □

Property · Double-angle formulas. For every real number a: $\sin 2a=2\sin a\cos a$ and $\cos 2a=\cos^2 a-\sin^2 a=2\cos^2 a-1=1-2\sin^2 a$.

Example 1 · An exact value. $\cos\frac{\pi}{12}=\cos\left(\frac{\pi}{3}-\frac{\pi}{4}\right)=\frac{1}{2}\times\frac{\sqrt{2}}{2}+\frac{\sqrt{3}}{2}\times\frac{\sqrt{2}}{2}=\frac{\sqrt{6}+\sqrt{2}}{4}$.

Product and arguments

Property · Product of two trigonometric forms. For all real numbers θ and θ': $(\cos\theta+i\sin\theta)(\cos\theta'+i\sin\theta')=\cos(\theta+\theta')+i\sin(\theta+\theta')$.

Consequently, for all nonzero complex numbers z, z' and every integer n:

$$\arg(zz')\equiv\arg z+\arg z', \qquad \arg\frac{1}{z}\equiv-\arg z,$$

$$\arg\frac{z}{z'}\equiv\arg z-\arg z', \qquad \arg(z^n)\equiv n\arg z \pmod{2\pi}.$$

Proof. Expanding gives $(\cos\theta\cos\theta'-\sin\theta\sin\theta')+i(\sin\theta\cos\theta'+\cos\theta\sin\theta')$, which the addition formulas simplify. If $z=r(\cos\theta+i\sin\theta)$ and $z'=r'(\cos\theta'+i\sin\theta')$, then $zz'=rr'(\cos(\theta+\theta')+i\sin(\theta+\theta'))$ with $rr'>0$: this is a trigonometric form of zz'. The other relations follow: $z\times\frac{1}{z}=1$ has argument 0, then $\frac{z}{z'}=z\times\frac{1}{z'}$, and an induction for $n\geq 0$, extended to $n<0$ through the inverse. □

Key point · Interpretation. Multiplying by $\cos\theta+i\sin\theta$ turns through an angle θ about O. The conjecture of the activity of chapter 2 is proved: multiplying by i is a direct quarter turn, multiplying by $1+i=\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$ is an eighth of a turn followed by an enlargement with ratio $\sqrt{2}$.

Exponential form

Definition · Imaginary exponential. For every real number θ, we write $e^{i\theta} = \cos\theta + i\sin\theta$. Every nonzero complex number can be written $z = re^{i\theta}$ with $r = |z|$ and $\theta \equiv \arg z \pmod{2\pi}$: this is its *exponential form*.

Property · Rules of computation. For all real numbers θ, θ' and every integer n:

$$e^{i\theta}e^{i\theta'} = e^{i(\theta+\theta')}, \qquad \frac{1}{e^{i\theta}} = e^{-i\theta} = \overline{e^{i\theta}}, \qquad \frac{e^{i\theta}}{e^{i\theta'}} = e^{i(\theta-\theta')}, \qquad \left(e^{i\theta}\right)^n = e^{in\theta}.$$

Values to know: $e^{i0} = 1$, $e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{-i\pi/2} = -i$, $\left|e^{i\theta}\right| = 1$.

Property · De Moivre's formula. For every real number θ and every integer n: $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$.

Property · Euler's formulas. For every real number θ: $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$ and $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$.

Proof. $e^{i\theta} + e^{-i\theta} = e^{i\theta} + \overline{e^{i\theta}} = 2\,\mathrm{Re}\left(e^{i\theta}\right) = 2\cos\theta$, and $e^{i\theta} - e^{-i\theta} = 2i\,\mathrm{Im}\left(e^{i\theta}\right) = 2i\sin\theta$. □

Method · Choosing the suitable form. For a sum, the algebraic form; for a product, a quotient or a power, the exponential form. Example: $1 + i\sqrt{3} = 2e^{i\pi/3}$, so

$$\left(1 + i\sqrt{3}\right)^{10} = 2^{10}e^{10i\pi/3} = 1{,}024\,e^{-2i\pi/3} = -512 - 512\sqrt{3}\,i,$$

since $\dfrac{10\pi}{3} - 4\pi = -\dfrac{2\pi}{3}$.

Method · Expressing $\cos 3\theta$ in terms of $\cos\theta$: de Moivre. Expand $(\cos\theta + i\sin\theta)^3$ with the binomial formula and identify the real part: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$.

Method · Linearising $\cos^3\theta$: Euler. $\cos^3\theta = \dfrac{\left(e^{i\theta} + e^{-i\theta}\right)^3}{8} = \dfrac{e^{3i\theta} + 3e^{i\theta} + 3e^{-i\theta} + e^{-3i\theta}}{8} = \dfrac{\cos 3\theta + 3\cos\theta}{4}$.

Method · Half angle. To factorise a sum of exponentials, take out the exponential of the mean angle: $e^{ia} + e^{ib} = e^{i\frac{a+b}{2}}\left(e^{i\frac{a-b}{2}} + e^{-i\frac{a-b}{2}}\right) = 2\cos\dfrac{a-b}{2}\,e^{i\frac{a+b}{2}}$.

Watch out. $e^{i\theta}$ is not a positive real number: its modulus is 1, but it can be negative ($e^{i\pi} = -1$) or non-real. An expression $re^{i\theta}$ is an exponential form only if $r > 0$.